A COUNTEREXAMPLE TO DIMENSION-TWO STABILIZATION FOR $\mathbf{M}_n(S_7)$

Abstract. Jiao and Ren asked whether $\mathsf{V}(\mathbf{M}_n(S_7)) = \mathsf{V}(\mathbf{M}_{n+1}(S_7))$ for every $n \geq 2$ and suggested that the chain may stabilize at dimension two. We show that this is false: the identity

$$x + y + x^2 + y^2 \approx x + y + x^2 + y^2 + yx$$

holds in $\mathbf{M}_2(S_7)$ and fails in $\mathbf{M}_3(S_7)$. Consequently, $\mathsf{V}(\mathbf{M}_2(S_7)) \subsetneq \mathsf{V}(\mathbf{M}_3(S_7))$. The same identity fails in $\mathbf{M}_n(S_7)$ for every $n \geq 3$.

An *additively idempotent semiring* is an algebra $(S, +, \cdot)$ whose additive reduct is a commutative idempotent semigroup, whose multiplicative reduct is a semigroup, and in which multiplication distributes over addition. For such an algebra A, let $\mathsf{V}(A)$ denote the variety it generates. Let $S_7 = \{1, a, \infty\}$ be the additively idempotent semiring with Cayley tables

$+$	1	a	∞
1	1	∞	∞
a	∞	a	∞
∞	∞	∞	∞

$\cdot$	1	a	∞
1	1	a	∞
a	a	∞	∞
∞	∞	∞	∞.

For $n \geq 2$, let $\mathbf{M}_n(S_7)$ be the semiring of $n \times n$ matrices over S_7. Jiao and Ren proved that $\mathbf{M}_n(S)$ embeds into $\mathbf{M}_{n+1}(S)$ for every additively idempotent semiring S and every $n \geq 2$. They asked whether the resulting chain for S_7 is constant from dimension two onward and observed that their nilpotency result suggests an affirmative answer [1, Theorem 2.1 and Section 5]. We show that the inclusion at $n = 2$ is strict.

Theorem 1. *The identity*

$$x + y + x^2 + y^2 \approx x + y + x^2 + y^2 + yx \tag{1}$$

holds in $\mathbf{M}_2(S_7)$ and fails in $\mathbf{M}_n(S_7)$ for every $n \geq 3$. Consequently,

$$\mathsf{V}(\mathbf{M}_2(S_7)) \subsetneq \mathsf{V}(\mathbf{M}_n(S_7)) \qquad (n \geq 3).$$

In particular, $\mathsf{V}(\mathbf{M}_2(S_7)) \subsetneq \mathsf{V}(\mathbf{M}_3(S_7))$.

Proof. Let $X, Y \in \mathbf{M}_2(S_7)$ and set

$$P = X + Y + X^2 + Y^2.$$

We prove $P + YX = P$ entrywise. From the Cayley tables, a finite nonempty sum in S_7 is 1 precisely when all of its summands are 1, and is a precisely

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when all of its summands are a. Also, a product is 1 only when both factors are 1, while $sa = as = a$ holds only for $s = 1$.

Fix $i, j \in \{1, 2\}$. If $P_{ij} = \infty$, then $(P + YX)_{ij} = P_{ij}$. Suppose $P_{ij} = 1$. Then $(X^2)_{ij} = (Y^2)_{ij} = 1$. Hence, for each $k \in \{1, 2\}$,

$$Y_{ik}Y_{kj} = 1 \quad \text{and} \quad X_{ik}X_{kj} = 1.$$

Thus $Y_{ik} = X_{kj} = 1$ for each k, so $(YX)_{ij} = 1$.

Suppose instead that $P_{ij} = a$. Then

$$X_{ij} = Y_{ij} = (X^2)_{ij} = (Y^2)_{ij} = a.$$

We have $i \neq j$, since otherwise the summand $X_{ii}^2 = a^2 = \infty$ would force $(X^2)_{ii} = \infty$. Since $i \neq j$ and the matrices are 2×2, the summation indices are precisely i and j, and therefore

$$a = (X^2)_{ij} = X_{ii}a + aX_{jj}.$$

Both summands must be a, so $X_{ii} = X_{jj} = 1$. The same argument gives $Y_{ii} = Y_{jj} = 1$. Consequently,

$$(YX)_{ij} = Y_{ii}X_{ij} + Y_{ij}X_{jj} = 1 \cdot a + a \cdot 1 = a.$$

This proves (1) in $\mathbf{M}_2(S_7)$.

For failure in dimension three, take

$$X = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ a & a & 1 \end{pmatrix}, \qquad Y = \begin{pmatrix} 1 & 1 & 1 \\ a & 1 & 1 \\ a & 1 & 1 \end{pmatrix}.$$

At entry $(3, 1)$,

$$X_{31} = Y_{31} = a, \qquad (X^2)_{31} = a \cdot 1 + a \cdot 1 + 1 \cdot a = a,$$

$$(Y^2)_{31} = a \cdot 1 + 1 \cdot a + 1 \cdot a = a, \qquad (YX)_{31} = a \cdot 1 + 1 \cdot 1 + 1 \cdot a = \infty.$$

Thus the two sides of (1) have respective $(3, 1)$ entries a and ∞, so the identity fails in $\mathbf{M}_3(S_7)$.

By the embedding theorem of Jiao and Ren, $\mathbf{M}_3(S_7)$ embeds into $\mathbf{M}_n(S_7)$ for every $n \geq 3$; hence (1) fails in every such $\mathbf{M}_n(S_7)$. Their theorem also gives $\mathbf{M}_2(S_7) \hookrightarrow \mathbf{M}_n(S_7)$, and therefore $\mathsf{V}(\mathbf{M}_2(S_7)) \subseteq \mathsf{V}(\mathbf{M}_n(S_7))$. Since an algebra and the variety it generates satisfy the same identities, (1) makes the inclusion strict. $\square$

Theorem 1 gives a negative answer to the question of Jiao and Ren and disproves stabilization at $\mathsf{V}(\mathbf{M}_2(S_7))$. It does not distinguish consecutive varieties from dimension three onward and therefore does not determine whether the chain stabilizes at a later dimension.

References

[1] J. Jiao and M. Ren, *The finite basis problem for matrix semirings* $\mathbf{M}_n(S_7)$, arXiv:2607.09677v1 [math.RA], 2026.